

Mechanics Lecture Notes

Kinetic Energy and Work

Kinetic Energy

Definition

Kinetic energy, denoted by K , is the form of energy possessed by a body due to its motion.

Key Features

- It is a **scalar quantity** that depends on the *mass* and *velocity* of the object.
- The faster an object moves, the greater its kinetic energy.
- If an object is at rest ($v = 0$), then its kinetic energy is **zero**.

Mathematical Expression

In classical mechanics, for a particle of mass m moving with velocity v (where $v \ll c$, c being the speed of light), the kinetic energy is:

$$K = \frac{1}{2}mv^2$$

Interpretation

- $K \propto m$ (heavier objects have more kinetic energy for the same speed).
- $K \propto v^2$ (if velocity doubles, kinetic energy becomes four times greater).

Significance

Kinetic energy is fundamental in analyzing motion, collisions, and energy conservation in mechanics. It helps explain how energy is transformed during work and interactions between objects.

Work

Definition

Work is defined as the product of force and displacement in the direction of the force.

General Expression

Mathematically, work is given by:

$$W = \mathbf{F} \cdot d,$$

where $\mathbf{F}$ is the force applied, and d is the displacement vector.

Special Case: Constant Force

For a constant force applied along a straight line, the work simplifies to:

$$W = Fd \cos \theta,$$

where

- F is the magnitude of the force,
- d is the displacement,
- θ is the angle between the force and displacement vectors.

Work and Angle Dependence

- When $\theta = 0^\circ$: Force is in the direction of displacement $\Rightarrow W = Fd$ (maximum work).
- When $\theta = 90^\circ$: Force is perpendicular to displacement $\Rightarrow W = 0$ (no work done).
- When $\theta = 180^\circ$: Force is opposite to displacement $\Rightarrow W = -Fd$ (negative work).

Thus, the angle θ plays a crucial role in determining how much work is done by the force.

The Work–Kinetic Energy Theorem

Statement

The **Work–Kinetic Energy Theorem** states that the change in kinetic energy of an object is equal to the net work done on it:

$$\Delta K = K_f - K_i = W.$$

It can also be written as:

$$K_f = K_i + W,$$

meaning:

$$\textit{Final kinetic energy} = \textit{Initial kinetic energy} + \textit{Net work done}.$$

Work–Energy Theorem (Proof)

A bead slides on a frictionless wire along the x -axis. A constant force $\mathbf{F}$ is applied on the bead which results in an acceleration a with the axis. So, From Newton's second law:

$$F_x = ma_x.$$

Multiply both sides by the displacement d :

$$F_x d = ma_x d.$$

From kinematics, since acceleration is constant,

$$v^2 - v_0^2 = 2a_x d \quad \Rightarrow \quad a_x d = \frac{v^2 - v_0^2}{2}.$$

Therefore,

$$F_x d = m \cdot \frac{v^2 - v_0^2}{2}.$$

Or,

$$F_x d = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2.$$

Thus, the work done equals the change in kinetic energy:

$$W = F_x d = \Delta K.$$

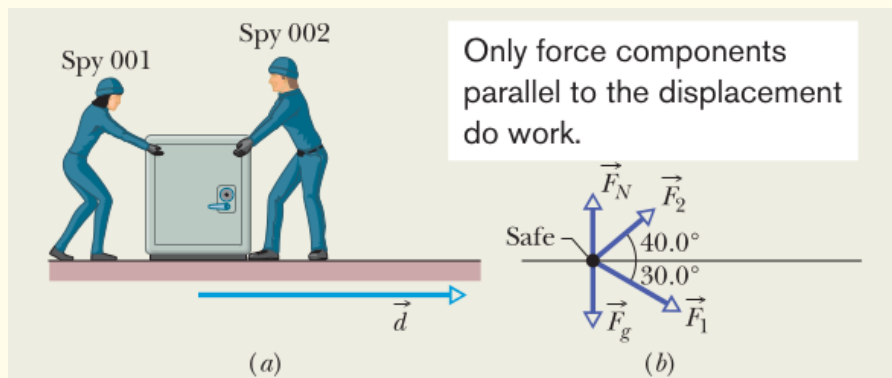
Problem 1: Sliding Safe

Problem Statement

Two industrial spies slide an initially stationary 225 kg floor safe a displacement $d = 8.50$ m.

- Spy 001 pushes with a force $F_1 = 12.0$ N at an angle 30.03° downward from the horizontal.
- Spy 002 pulls with a force $F_2 = 10.0$ N at an angle 40.03° above the horizontal.

The floor and safe are frictionless.



Questions:

- What is the net work done on the safe by F_1 and F_2 ?
- What is the work done by gravity W_g and by the normal force W_N ?
- If initially stationary, what is the final speed v_f of the safe?

Solution

(a) Net Work Done by F_1 and F_2

The horizontal components of the forces contribute to the work done on the safe. The horizontal components are:

$$F_{1x} = F_1 \cos 30.03^\circ = 12.0 \times \cos 30.03^\circ = 10.39 \text{ N}$$

$$F_{2x} = F_2 \cos 40.03^\circ = 10.0 \times \cos 40.03^\circ = 7.66 \text{ N}$$

The net force in the horizontal direction is:

$$F_{\text{net}} = F_{1x} + F_{2x} = 10.39 + 7.66 = 18.05 \text{ N}$$

The work done is given by:

$$W = F_{\text{net}}d = (18.05)(8.50) = 153.4 \text{ J}$$

(b) Work Done by Gravity and Normal Force

The gravitational force is:

$$F_g = mg = (225)(9.81) = 2207.25 \text{ N}$$

Since gravity acts perpendicular to the displacement, the work done by gravity is:

$$W_g = F_g d \cos 90^\circ = 0 \text{ J}$$

Similarly, the normal force is perpendicular to the displacement, so:

$$W_N = 0 \text{ J}$$

(c) Final Speed of the Safe

Using the work-energy theorem:

$$W_{\text{net}} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

Since the safe is initially stationary, $v_i = 0$, so:

$$v_f = \sqrt{\frac{2W}{m}} = \sqrt{\frac{2(153.4)}{225}}$$

$$v_f = \sqrt{1.363} = 1.17 \text{ m/s}$$

So, the final speed of the safe is 1.17 m/s.

Problem 2: Crate on a Slick Parking Lot

Problem Statement

During a storm, a crate of crepe is sliding across a slick, oily parking lot through a displacement:

$$\mathbf{d} = (-3.0 \text{ m})\hat{i},$$

while a steady wind pushes against the crate with a force:

$$\mathbf{F} = (2.0 \text{ N})\hat{i} + (-6.0 \text{ N})\hat{j}.$$

- (a) How much work does this force do on the crate during the displacement?
- (b) If the crate has a kinetic energy of 10 J at the beginning of displacement $\mathbf{d}$, what is its kinetic energy at the end of $\mathbf{d}$?

Solution

(a) Work Done by the Force

The work done by a force during a displacement is given by the dot product:

$$W = \mathbf{F} \cdot \mathbf{d}.$$

Expanding the dot product:

$$W = (2.0\hat{i} + (-6.0)\hat{j}) \cdot (-3.0\hat{i}).$$

Since $\hat{i} \cdot \hat{i} = 1$ and $\hat{j} \cdot \hat{i} = 0$, we get:

$$W = (2.0)(-3.0) + (-6.0)(0) = -6.0 \text{ J}.$$

Thus, the force does -6.0 J of work on the crate, meaning it removes energy from the system.

(b) Final Kinetic Energy

Using the work-energy theorem:

$$K_f = K_i + W.$$

Given $K_i = 10 \text{ J}$:

$$K_f = 10 + (-6.0) = 4.0 \text{ J}.$$

Thus, the kinetic energy of the crate at the end of the displacement is 4.0 J .

Work Done by Gravitational Force

The work W_g done by the gravitational force $\mathbf{F}_g$ on a particle-like object of mass m as the object moves through a displacement $\mathbf{d}$ is given by:

$$W_g = mgd \cos \theta,$$

where θ is the angle between $\mathbf{F}_g$ and $\mathbf{d}$.

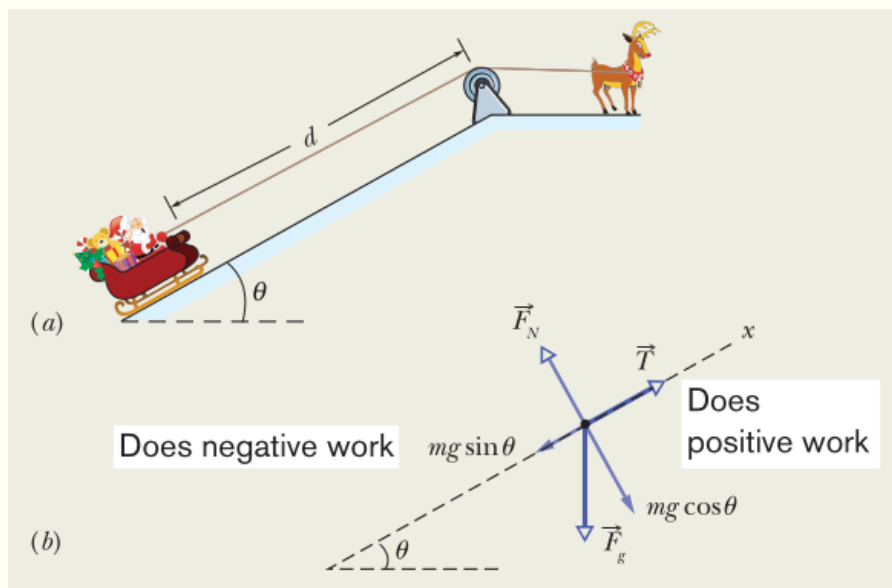
Problem 3: Work on a Sleigh

Problem Statement

An object is pulled along a ramp, but the object starts and ends at rest.

A rope pulls a 200 kg sleigh up a slope at an incline angle $\theta = 30^\circ$, through a distance $d = 20$ m. The sleigh and its contents have a total mass of 200 kg. The snowy slope is so slippery that we take it to be frictionless.

Question: How much work is done by each force acting on the sleigh?



Solution

Work Done by the Normal Force

The work W_N done by the normal force is zero since the normal force is perpendicular to the displacement of the sleigh:

$$W_N = F_N d \cos 90^\circ = 0$$

Work Done by the Gravitational Force

The angle between the gravitational force $\mathbf{F}_g$ and the displacement $\mathbf{d}$ is 120° (sum of the incline angle 30° and 90°):

$$W_g = F_g d \cos 120^\circ = mgd \cos 120^\circ$$

Substituting the values:

$$W_g = (200 \text{ kg})(9.8 \text{ m/s}^2)(20 \text{ m}) \cos 120^\circ$$

$$W_g = -1.96 \times 10^4 \text{ J}$$

Work Done by the Tension Force

The total work done on the sleigh is:

$$W = W_N + W_g + W_T$$

Since the sleigh starts and ends at rest, the change in kinetic energy is zero. By the work-energy theorem:

$$0 = W_N + W_g + W_T$$

Substituting the known values:

$$0 = 0 + (-1.96 \times 10^4) + W_T$$

Solving for W_T :

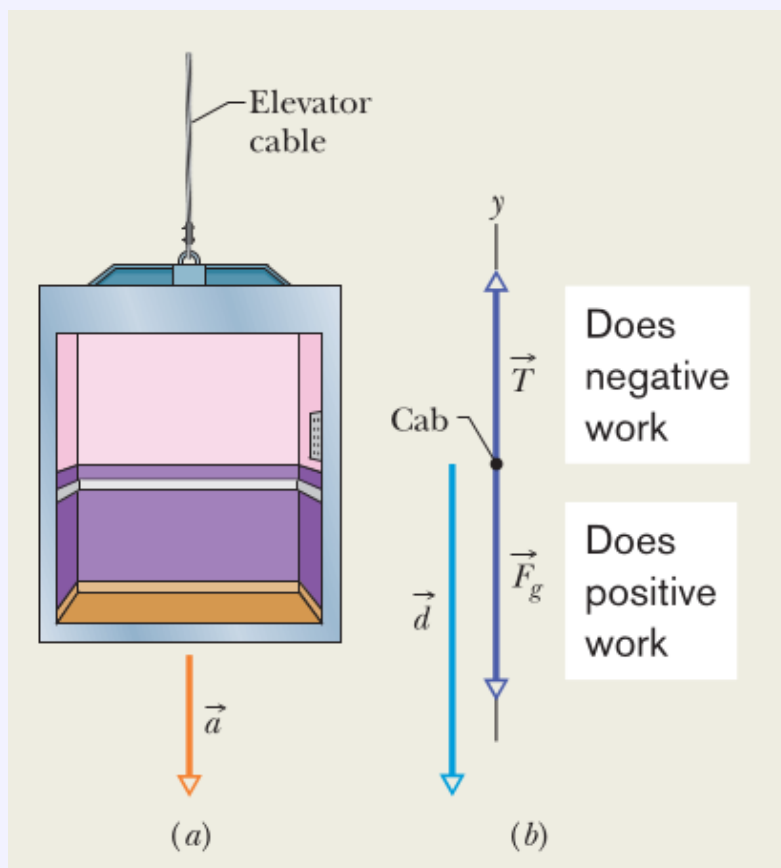
$$W_T = 1.96 \times 10^4 \text{ J}$$

Problem 4: Elevator Cab

Problem Statement

An elevator cab of mass $m = 500$ kg is descending with speed $v_i = 4.0$ m/s when its supporting cable begins to slip, allowing it to fall with constant acceleration $a = \frac{g}{5}$.

- (a) During the fall through a distance $d = 12$ m, what is the work W_g done on the cab by the gravitational force F_g ?
- (b) During the 12 m fall, what is the work W_T done by the tension in the cable?
- (c) What is the net work W done on the cab during the fall?
- (d) What is the cab's kinetic energy at the end of the 12 m fall?



Solution:

(a) Work done by the gravitational force

The angle between the gravitational force and the cab's displacement is 0° , so the work is

$$W_g = F_g \cdot d \cdot \cos 0^\circ$$

Since $F_g = mg$, we have

$$W_g = mgd \cdot 1 = (500 \text{ kg})(9.8 \text{ m/s}^2)(12 \text{ m}) = 5.88 \times 10^4 \text{ J} = 59 \text{ kJ}.$$

(b) Work done by the tension

From Newton's second law along the vertical direction:

$$T - F_g = -ma$$

Solving for the tension:

$$T = m(g - a)$$

The work done by tension is

$$W_T = Td \cos 180^\circ$$

Substituting $a = g/5$ downward:

$$W_T = m \left(g - \frac{g}{5} \right) d \cdot (-1) = -\frac{4mgd}{5}$$

Plugging in values:

$$W_T = -(500)(9.8)(12)\frac{4}{5} = -4.70 \times 10^4 \text{ J} = -47 \text{ kJ}.$$

(c) Net work on the cab

$$W = W_g + W_T = 5.88 \times 10^4 - 4.70 \times 10^4 = 1.18 \times 10^4 \text{ J} = 12 \text{ kJ}$$

(d) Final kinetic energy

The initial kinetic energy:

$$K_i = \frac{1}{2}mv_i^2 = \frac{1}{2}(500)(4.0)^2 = 4000 \text{ J} = 4.0 \text{ kJ}$$

Using the work-energy theorem:

$$K_f = K_i + W = 4.0 \text{ kJ} + 12 \text{ kJ} = 16 \text{ kJ}$$

Thus, the cab's final kinetic energy is $K_f = 1.58 \times 10^4 \text{ J} = 16 \text{ kJ}$.

Problem 5

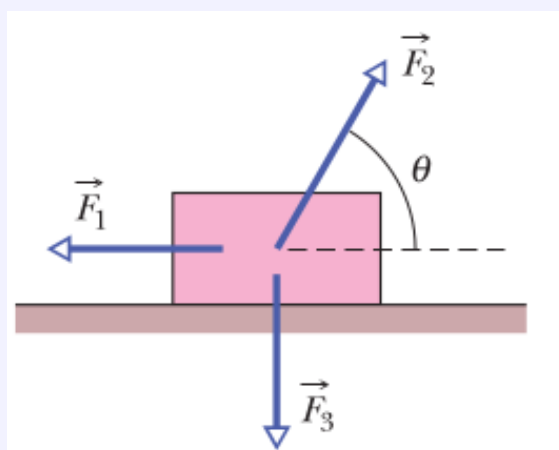
Problem Statement:

Figure shows three forces applied to a trunk that moves leftward by 3.00 m over a frictionless floor.

The force magnitudes are:

$$F_1 = 5.00 \text{ N}, \quad F_2 = 9.00 \text{ N}, \quad F_3 = 3.00 \text{ N},$$

and the indicated angle is: $\theta = 60.0^\circ$.



During the displacement:

- (a) What is the net work done on the trunk by the three forces?
- (b) Does the kinetic energy of the trunk increase or decrease?

Solution

The forces are constant, so the work done by any one of them is given by

$$W = \vec{F} \cdot \vec{d},$$

where $\vec{d}$ is the displacement.

Force $\vec{F}_1$ is in the direction of the displacement, so

$$W_1 = F_1 d \cos \phi_1 = (5.00 \text{ N})(3.00 \text{ m}) \cos 0^\circ = 15.0 \text{ J}.$$

Force $\vec{F}_2$ makes an angle of 120° with the displacement, so

$$W_2 = F_2 d \cos \phi_2 = (9.00 \text{ N})(3.00 \text{ m}) \cos 120^\circ = -13.5 \text{ J}.$$

Force $\vec{F}_3$ is perpendicular to the displacement, so

$$W_3 = F_3 d \cos \phi_3 = 0 \quad (\text{since } \cos 90^\circ = 0).$$

The net work done by the three forces is

$$W = W_1 + W_2 + W_3 = 15.0 \text{ J} - 13.5 \text{ J} + 0 = 1.50 \text{ J}.$$

(b) If no other forces do work on the trunk, its kinetic energy increases by 1.50 J during the displacement.